Sync_FIFO_SBRX

Black-Box Verification Specification — Clear Edition

A readable external-behavior specification for verification without requiring knowledge of the RTL implementation.

Block type	Single-clock synchronous FIFO
DATA_WIDTH default	64
FIFO_DEPTH default	64 entries
Clock	lclk; operations are evaluated on the rising edge
Reset	sys_rst; active-low asynchronous reset

Important interface note

The supplied RTL declares wr_data and rd_data as [DATA_WIDTH:0]. That means each port is DATA_WIDTH + 1 bits wide (65 bits with DATA_WIDTH=64). Verify this width with the RTL owner; do not silently assume 64 bits.

Document sections

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1. Interface Definition

Parameters

Parameter	Default	Meaning
DATA_WIDTH	64	Nominal width parameter. The declared ports [DATA_WIDTH:0] are DATA_WIDTH+1 bits wide.
FIFO_DEPTH	64	Number of entries intended to be stored.

Ports

Port	Dir.	Width	Description
lclk	Input	1 bit	Clock; requests are evaluated on rising edges.
sys_rst	Input	1 bit	Active-low asynchronous reset. Low resets FIFO state.
wr_en	Input	1 bit	Write request enable.
rd_en	Input	1 bit	Read request enable.
wr_data	Input	DATA_WIDTH+1	Data presented for a write request.
rd_data	Output	DATA_WIDTH+1	Most recent accepted read value. Holds when no read is accepted; reset sets it to zero.
full	Output	1 bit	High means a write cannot be accepted.
empty	Output	1 bit	High means no data is available to read.

2. Functional Behavior

The FIFO stores accepted writes in arrival order and returns accepted reads in the same order (first-in, first-out). Operations are evaluated at the rising edge of lclk when sys_rst is high.

Reset: when sys_rst is low, FIFO state is cleared asynchronously and rd_data becomes zero. After reset, the FIFO should be empty.

Write: accept a write only when wr_en=1 and full=0. The input wr_data is appended to the FIFO. When full=1, the write is ignored.

Read: accept a read only when rd_en=1 and empty=0. rd_data updates to the oldest stored entry. When empty=1, the read is ignored and rd_data holds its previous value.

Flags: empty indicates zero stored entries; full indicates FIFO_DEPTH stored entries. A successful write increases occupancy by one, a successful read decreases it by one, and a simultaneous accepted read/write keeps occupancy unchanged.

3. Operation Acceptance Table

wr_en	rd_en	Pre-edge condition	Expected result
0	0	Any state	No operation; rd_data holds.
1	0	full=0	Write accepted; occupancy +1.
1	0	full=1	Write rejected; contents unchanged.
0	1	empty=0	Read accepted; oldest item appears on rd_data; occupancy -1.
0	1	empty=1	Read rejected; rd_data holds.
1	1	full=0, empty=0	Read oldest pre-edge item and append new item; occupancy unchanged.
1	1	empty=1	Read rejected; write accepted if full=0.
1	1	full=1	Write rejected; read accepted if empty=0.

Boundary note: the behavior above follows the RTL's independent gating of writes by full and reads by empty. Confirm this contract with the design owner, especially for simultaneous requests at empty/full boundaries.

4. Required Verification Scenarios

Reset and initialization

Assert reset while idle and during traffic. Check rd_data=0 and empty=1; confirm pre-reset data is not returned after reset.

Single write then single read

Write a known pattern, check empty deasserts, read it back, compare data, and check empty asserts again.

FIFO ordering

Write distinct values and read all of them. Verify exact order with no loss, duplication, or reordering.

Fill to capacity

Write FIFO_DEPTH items. Check full asserts and all accepted entries are returned in order.

Write while full

Keep wr_en high while full=1. Confirm no extra write is accepted and existing contents stay unchanged.

Drain to empty

Read all valid entries; check every value and confirm empty asserts after the last item.

Read while empty

Assert rd_en while empty=1. Confirm no read is accepted and rd_data holds its previous value.

Simultaneous read/write

Test while partially occupied. Check the oldest item is read and the new item is appended.

Boundary transitions

Check status and data around occupancy 0, 1, FIFO_DEPTH-1, and FIFO_DEPTH.

Data patterns

Use zero, all ones, alternating bits, walking-one/walking-zero, and randomized data.

Long randomized traffic

Randomize enables and data; compare every accepted read against a software queue and continuously check flags.

Reset during traffic

Reset with data stored, then confirm all old items are discarded and operation restarts empty.

5. Reference-Model Guidance

Use a software queue as the golden model. Update it only for operations accepted according to full and empty sampled immediately before the rising edge.

1. On reset assertion, clear the reference queue and set expected rd_data to zero.

2. Before each rising edge, sample wr_en, rd_en, full, empty, and wr_data.

3. If rd_en=1 and empty=0, the expected read value is the queue's oldest item.

4. If wr_en=1 and full=0, append wr_data to the queue.

5. For simultaneous read/write, determine the read value from the pre-edge queue, then apply the accepted write.

6. Check rd_data after the edge only for an accepted read; otherwise verify it holds, except during reset.

7. After outputs settle, compare full and empty with queue occupancy.

Core checks

• empty=1 whenever reference occupancy is zero.

• full=1 whenever reference occupancy equals FIFO_DEPTH.

• Reference occupancy stays between zero and FIFO_DEPTH.

• Every accepted read returns the oldest queued item.

• A rejected read does not change rd_data.

• A rejected write does not change stored sequence.

• Accepted writes minus accepted reads equals current occupancy since reset.

6. Assumptions and Open Points

Confirm these points with the RTL owner before sign-off:

Data width: [DATA_WIDTH:0] means DATA_WIDTH+1 bits. Confirm whether this is intentional or should be [DATA_WIDTH-1:0].

FIFO_DEPTH: confirm whether it must be a power of two. Non-power-of-two depths require clearly defined pointer wraparound behavior.

Simultaneous operations at boundaries: confirm expected read/write acceptance when both enables are high and the FIFO is empty or full.

Flag timing: full and empty are status outputs; sample them after simulation has settled following the active edge.

> Sign-off criterion: all directed and randomized tests pass; no accepted data is lost, duplicated, or reordered; boundary behavior matches the agreed contract; and full/empty match the reference occupancy.